

نماذج الاختبارات الشهرية

تجميع :

نور الجفري & فاطمة عاشور

Question (1): (8+4+3=15 marks)

a) Determine whether each of the following is true or false and correct the false statement:

- $\ln \sqrt[3]{e} = 3$ $\ln \sqrt[3]{e} = \ln e^{\frac{1}{3}} = \frac{1}{3}$ (... False ...)
- If $A(1, -1), B(-3, 2)$ is $d(A, B) = 5$ (... True ...)
- If $f(x) = \frac{x^2+5}{x+6}$ then $D_f = \mathbb{R}$ $D_f = \mathbb{R} \setminus \{-6\}$ (... False ...)
- The equation of the circle with center $(3, -2)$ and radius 4 is $(x-3)^2 + (y-2)^2 = 16$ (... False ...)
 $(x-3)^2 + (y+2)^2 = 16$

b) Find the equation of a line passes through the points $(5, -7)$ that is perpendicular to the line $\vec{l}_2 : 6x + 3y = 4$.

solution:

$$m_2 = -\frac{6}{3} = -2$$

$$\therefore L_1 \perp L_2 \Rightarrow m_1 = -\frac{1}{m_2} = -\frac{1}{-2} = \frac{1}{2}$$

The equation is:

$$y+7 = \frac{1}{2}(x-5) \Rightarrow 2y+14 = x-5 \Rightarrow 2y+14-x+5=0$$

$$\Rightarrow 2y - x + 19 = 0$$

$$\frac{4}{21}$$

c) verify the identity: $\tan^2 \theta (1 + \cot^2 \theta) = \frac{1}{1 - \sin^2 \theta}$

solution:

$$\tan^2 \theta (1 + \cot^2 \theta) = \tan^2 \theta \cdot \csc^2 \theta = \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \frac{1}{\sin^2 \theta} = \frac{1}{\cos^2 \theta} = \frac{1}{1 - \sin^2 \theta}$$

$$\Rightarrow \frac{1}{1 - \sin^2 \theta}$$

Question (2) : (8+4+3=15 marks)

a) Choose the correct answer

1) $[-2] = \underline{-2}$

(a) -1 , (b) ~~-2~~ , (c) -9

2) $f(x) = \sin x$ is ~~odd~~ function (a) even , (b) ~~odd~~ , (c) neither even nor odd

3) The set of solution of the inequality $\frac{x+1}{x-4} < 0$ is ~~$(-1, 4)$~~

(a) $(-\infty, -1) \cup (4, \infty)$, (b) $[-1, 4]$, (c) ~~$(-1, 4)$~~

4) If $\theta = \frac{\pi}{2}$, the equation of circle is $(x-1)^2 + (y+1)^2 = 4$ then the area of a sector is

$A = \dots \dots \pi \dots \dots$ (a) $\frac{2\pi}{4}$, (b) $\frac{3\pi}{2}$, (c) ~~π~~

$\frac{8}{8}$

b) Let $f(x) = x^2$, $g(x) = x - 1$, find $(f \circ g)(x)$ and sketch the graph of $f \circ g$

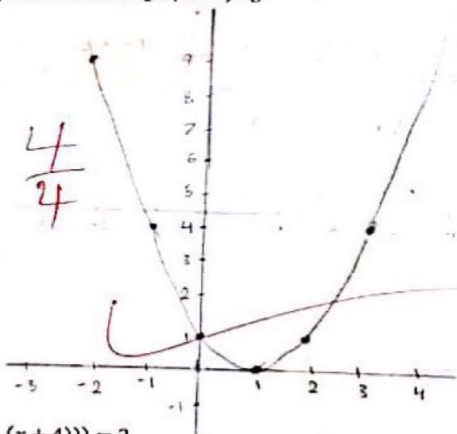
solution:-

$(f \circ g)(x) = f(g(x)) = f(x-1) = (x-1)^2$

$= x^2 - 2x + 1$

$D_{f \circ g} = \mathbb{R}$

x	-1	-2	0	1	2	3
$(f \circ g)(x)$	4	9	1	0	1	4



$\frac{4}{2}$

c) Solve the equation : $\log_3(7 + \log_4(13 + \log_2(x+4))) = 2$

solution:-

$3^2 = 9 = 7 + \log_4(13 + \log_2(x+4)) \Rightarrow \log_4(13 + \log_2(x+4)) = 9 - 7 = 2$

$\Rightarrow 4^2 = 16 = 13 + \log_2(x+4) \Rightarrow \log_2(x+4) = 16 - 13 = 3$

~~2~~ 3 3 3

$\Rightarrow 2^3 = 8 = x + 4 \Rightarrow x = 8 - 4 = 4$

$\Rightarrow x = 4$

$$\log_3 (7 + \log_4 (13 + \log_2 (x+4))) = 2 \quad (i)$$

solution

$$3^2 = 9 = 7 + \log_4 (13 + \log_2 (x+4)) \Rightarrow \log_4 (13 + \log_2 (x+4)) = 9-7 = 2$$

$$\Rightarrow 4^2 = 16 = 13 + \log_2 (x+4) \Rightarrow \log_2 (x+4) = 16-13 = 3$$

$$\Rightarrow 2^3 = 8 = x+4 \Rightarrow x = 8-4 = 4$$

$$\Rightarrow \boxed{x = 4}$$

Question (1): (8+4+3=15 marks)

a) Determine whether each of the following is true or false and correct the false statement:

1) $e^{5 \ln 2} = 10$

2) If $A(1,0), B(-3,2)$ is $M(A,B) = (-1,0)$

3) If $f(x) = \frac{x}{x-1}$ then $D_f = \mathbb{R} \setminus \{-1\}$

4) The equation of the circle with center $(-2,3)$ and radius 5 is $(x+2)^2 + (y-3)^2 = 25$

b) Find the equation of a line passes through the points $(5,-7)$ that is parallel to the line

$L_2: 6x + 3y = 4$

$y - y_1 = m(x - x_1)$
 $y + 7 = \frac{1}{2}(x - 5)$

$7 + y = \frac{1}{2}x + \frac{5}{2}$
 $= \frac{1}{2}x - y + \frac{5}{2} + 7$
 $-x - 2y + 5 - 14$
 $-x - 2y - 9 = 0$

$\sin^2 \theta + \cos^2 \theta = 1$

c) verify the identity: $\frac{\sin^2 \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$

$\frac{\sin \theta}{1 + \cos \theta} \times \frac{1 - \cos \theta}{1 - \cos \theta} = \frac{\sin \theta (1 - \cos \theta)}{1 - \cos^2 \theta}$

$= \frac{\sin \theta (1 - \cos \theta)}{\sin^2 \theta} = \frac{1 - \cos \theta}{\sin \theta}$

$\sin^2 \theta + \cos^2 \theta = 1$
 $\sin^2 \theta = 1 - \cos^2 \theta$

Question (2): (8+4+3=15 marks)

a) Choose the correct answer

1) $[-1] =$

(a) -1, (b) -2, (c) 0

2) $f(x) = \cos x$ is ... function (a) even, (b) odd, (c) neither even nor odd

3) The set of solution of the inequality $\frac{x-3}{x-1} < 0$ is

(a) $(-\infty, 1) \cup (3, \infty)$, (b) $[1, 3]$, (c) $(1, 3)$

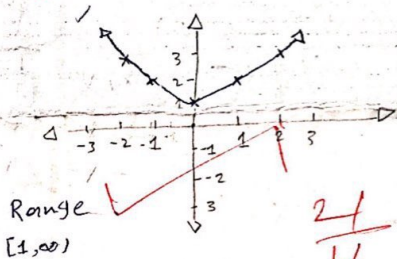
4) If $\theta = \frac{\pi}{17}$, endpoints of a diameter A(4, -3), B(-2, 3) then $S =$

(a) 2π , (b) $\frac{2\pi}{3}$, (c) $\pi \sqrt{\frac{1}{5}}$

b) Let $f(x) = x^2$, $g(x) = x + 1$, find $(g \circ f)(x)$ and sketch the graph of $g \circ f$

$(g \circ f)(x) = g(f(x)) = x^2 + 1$

X	-2	-1	0	1	2
f(x)	3	2	1	2	3



c) Solve the equation: $\log_4 x + \log_{16} x + \log_2 x = 7$

$\frac{7}{4} = x$, $\frac{7}{16} = x$, $\frac{7}{2} = x$

16384

$\log_4 x + \log_{16} x + \log_2 x = 7$

$\log_2 x + \log_2 x + \log_2 x = 7$

$2 \log_2 x + \log_2 x + \log_2 x = 7$

$(\log_2 x)^4 = 7 \Rightarrow x = 16384$

$\log_2 x = 7 \Rightarrow x = 128$

$\log_2 x = 7 \Rightarrow x = 128$

0/3

$$\log_4 x + \log_{16} x + \log_2 x = 7$$

$$\frac{\log_2 x}{\log_2 2} + \frac{\log_2 x}{\log_2 2^2} + \frac{\log_2 x}{\log_2 2} = 7 \Rightarrow$$

$$\frac{\log_2 x}{1} + \frac{\log_2 x}{2} + \frac{\log_2 x}{1} = 7$$

$$4 \log_2 x + 2 \log_2 x + \log_2 x = 7$$

$$7 \log_2 x = 7 \Rightarrow \log_2 x = 1$$

$$\boxed{x = 2^1 = 2}$$

EMKO

Name: _____

Department: IT

Q1) a) Determine whether each of the following is true or false:

1) $\lim_{x \rightarrow 1} (x^2 - 2x + 5) = 4$

2) $\cot \frac{7\pi}{4} = -1$

3) If $6^{2x \log_6 5} = 125$ then $x = \frac{3}{2}$

(...false...)

(...false...)

(...True...)

b) 1) Given the ellipse equation, find the eccentricity, the coordinates of the vertices, the foci, the equations of the directrices, and sketch the graph of $\frac{x^2}{16} + \frac{y^2}{25} = 1$

$a^2 = 25$

$b^2 = 16$

$c^2 = a^2 - b^2$

$c^2 = 25 - 16$

$c^2 = 9$

$c = 3$

vertices: $(0, \pm 5)$

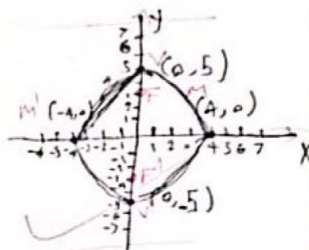
foci: $(0, \pm 3)$

$e = \frac{c}{a}$

$e = \frac{3}{5}$

directrices: $y = \pm \frac{a^2}{c}$

directrices: $y = \pm \frac{25}{3} = \pm 8.33$



2) Show that $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}$

$\frac{\sin \theta}{1 - \cos \theta} = \frac{\sin \theta}{1 - \cos \theta}$

$\frac{\sin \theta}{1 - \cos \theta} = \frac{\sin \theta}{1 - \cos^2 \theta}$

$\frac{\sin \theta}{1 - \cos \theta} = \frac{\sin \theta}{\sqrt{1 - \cos^2 \theta}}$

$\frac{\sin \theta}{1 - \cos \theta} = \frac{\sin \theta}{1 - \cos \theta}$

$1 + \tan^2 \theta = \sec^2 \theta$

$\sin^2 \theta + \cos^2 \theta = 1$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$

$\sin \theta = \frac{1}{\csc \theta}$

$\csc \theta = \frac{1}{\sin \theta}$

$\tan \theta = \frac{1}{\cot \theta}$
 $\cot \theta = \frac{1}{\tan \theta}$

$\cot \theta = \frac{\cos \theta}{\sin \theta}$

$\cos \theta = \frac{1}{\sec \theta}$

$\sec \theta = \frac{1}{\cos \theta}$

Q2) a) Choose the correct answer

1) If $\tan \theta = \frac{3}{4}$ and $\theta \in III$, then $\sec \theta =$

$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\frac{4}{5}} = \frac{5}{4}$
 $\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\frac{4}{5}} = \frac{5}{4}$
 $\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\frac{4}{5}} = \frac{5}{4}$

2) If $\frac{1}{b^1} = 3^b$, then $b =$

(a) -3

(b) $\frac{1}{3}$

(c) $\frac{-1}{2}$

3) $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x} =$

(a) 4

(b) $\frac{4}{3}$

(c) $-\frac{4}{3}$

b) 1) Evaluate $\lim_{k \rightarrow 6} \frac{k^2 - 36}{k - 6}$ (if they exist)

$\frac{k^2 - 36}{k - 6}$

$\frac{6^2 - 36}{6 - 6} = \frac{0}{0}$

$\lim_{k \rightarrow 6} \frac{k^2 - 36}{k - 6} = \frac{(k-6)(k+6)}{k-6}$

$\lim_{k \rightarrow 6} (k+6)$

$6+6$

$= 12$

$\frac{2}{2}$

2) Find $\log_{100} 25 + \log_{100} 4$ (without the calculator)

$\log_{100} 25 + \log_{100} 4$

$(\log_{100} 25)(\log_{100} 4) = 1.0$

$\log_{100} 100$

$\log_{100} 2$

$\log_{100} 2$

$= 1$

$\frac{2}{2}$

Question (1): (8+4+3=15 marks)

a) Determine whether each of the following is true or false and correct the false statement:

1) If $2^{x+2} = 8^{2x}$ then $x = \frac{1}{8}$ $2^{x+2} = 2^{3 \cdot 2x} \Rightarrow 2^{x+2} = 2^{6x} \Rightarrow x+2 = 6x \Rightarrow 2 = 5x \Rightarrow x = \frac{2}{5}$ (False, ...)

2) The slope of a line $y = 4$ is 0 $\frac{1}{2} = \frac{5}{5} \Rightarrow x = \frac{1}{5}$ (True, ...)

3) If $f(x) = x + 1$, $g(x) = \sqrt{x^2 - 1}$ then $(f \circ g)(x) = \sqrt{x+1} + \sqrt{x-1} + 1$ (False, ...)

4) The equation of the circle that satisfying the conditions: endpoints of diameter

$A(4, -3), B(-2, 7)$ is $(x-1)^2 + (y-2)^2 = 29$ (False, ...)

b) Find the equation of a line passes through the points $A(1, 7), B(-3, 2)$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 7}{-3 - 1} = \frac{-5}{-4} = \frac{5}{4}$

$y = y_1 + m(x - x_1) \Rightarrow y = 7 + \frac{5}{4}(x - 1) \Rightarrow y = 7 + \frac{5x}{4} - \frac{5}{4}$

$y = 7 + \frac{5x - 5}{4}$

$4y = 28 + 5x - 5$

$4y = 28 + 5x - 5$

$4y - 5x - 23 = 0$

$4y - 5x - 23 = 0$

c) If $\csc \theta = \frac{\sqrt{13}}{2}$ and $\theta \in III$. Find the value of the other trigonometric function.

$\sin \theta = \frac{1}{\csc \theta}$

$\sin \theta = \frac{1}{\frac{\sqrt{13}}{2}} = \frac{2}{\sqrt{13}}$

$\cos^2 \theta = 1 - \sin^2 \theta$

$\cos^2 \theta = 1 - \left(\frac{2}{\sqrt{13}}\right)^2$

$\cos^2 \theta = 1 - \frac{4}{13}$

$\sqrt{\cos^2 \theta} = \frac{13-4}{13} = \frac{9}{13} = \sqrt{\frac{9}{13}}$

$\cos \theta = \frac{-\sqrt{9}}{\sqrt{13}} = \frac{-3}{\sqrt{13}}$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$

$\tan \theta = \frac{\frac{2}{\sqrt{13}}}{\frac{-3}{\sqrt{13}}} = \frac{-2}{3}$

$\tan \theta = \frac{-2}{\sqrt{13}} \times \frac{\sqrt{13}}{-3} = \frac{-2}{-3} = \frac{2}{3}$

$\cot \theta = \frac{1}{\tan \theta} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$

$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\frac{-3}{\sqrt{13}}} = \frac{-\sqrt{13}}{3}$

$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\frac{-3}{\sqrt{13}}} = \frac{-\sqrt{13}}{3}$

$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\frac{-3}{\sqrt{13}}} = \frac{-\sqrt{13}}{3}$

Question (2) : (8+3+3=15 marks)

a) Choose the correct answer

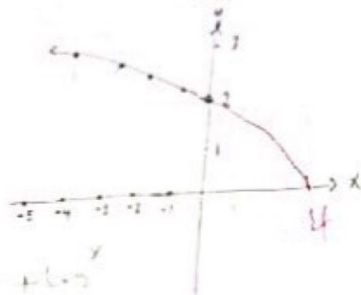
- 1) $[1.9] = \dots\dots\dots$ 1
- 2) $f(x) = |x| + 5$ is $\dots\dots\dots$ function (a) even, (b) odd, (c) neither even nor odd
- 3) The set of solution of the inequality $x^2 - 3x - 4 \leq 0$ is $\dots\dots\dots$
(a) $(-\infty, -1] \cup [4, \infty)$, (b) $[-1, 4]$, (c) $(-1, 4)$
- 4) If $\theta = \frac{3\pi}{2}$, $r = 2\text{cm}$ then $A = \dots\dots\dots$ (a) 3π , (b) $\frac{3\pi}{2}$, (c) π

b) Let $f(x) = \sqrt{4-x}$, find D_f and sketch the graph of f

$$D_f: 4-x \geq 0 \Rightarrow -x \geq -4 \Rightarrow x \leq 4 \quad [-4, \infty) \quad (-\infty, 4]$$

x	-4	-3	-2	-1	0
f(x)	$2\sqrt{2}$	$\sqrt{7}$	$\sqrt{6}$	$\sqrt{5}$	2

$$R_f: [2, \infty)$$



c) Find the value of $\log_a \left(\frac{y}{x}\right)^3 + \log_a x^2 - \log_a y^2 + \log_a \frac{x}{y}$ where $a, x, y \in \mathbb{R}^+$, $a \neq 1$

$$\log_a \frac{y^3}{x^3} - \log_a \frac{y^2}{x^2}$$

$$\log_a \frac{y^3 x^2}{x^3 y^2}$$

$$\log_a \frac{y}{x}$$

$$\log_a \frac{y}{x}$$

$$\log_a \frac{y}{x}$$

$$a=2$$

$$y=3$$

$$x=4$$

$$\log_a \frac{y}{x}$$

$$\log_a \frac{y}{x}$$

$$\log_a \frac{y}{x}$$

$$\log_a \frac{y}{x}$$

$$\log_a \frac{y}{x}$$

$$\log_a \frac{y}{x}$$

Name: _____

Department: CS

Question (1): (8+4+3=15 marks)

a) Determine whether each of the following is **true** or **false** and correct the false statement:

1) If $3^{2x-5} = \frac{1}{9}$ then $x = \frac{5}{2}$

$\frac{6}{8}$

(...true...)

2) Let $A(2, -1), B(-3, 4)$ then $d(A, B) = \sqrt{50}$

$\frac{6}{8}$

(...true...)

3) If $(f \circ g)(x) = \sqrt{x-1}$, then $(f \circ g)(5) = (4) = \sqrt{4} = 2$

(...false...)

4) The equation of the circle that satisfying the conditions: center $(-4, 2)$, tangent to the

x-axis is $(x+4)^2 + (y-2)^2 = 16$

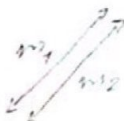
$\frac{4}{4}$

(...true...)

b) Find the equation of a line ($\vec{L}_1$) passes through the point $(-3, 2)$ that is parallel to the line $\vec{L}_2: x + 3y - 3 = 0$.

$y - y_1 = m(x - x_1)$

$x + 3y - 3 = 0$



$3y = -x + 3$

$y = \frac{-x + 3}{3}$

$y = \frac{-x}{3} + 1$

$m_2 = \frac{-1}{3} = m_1$

$m_2 = \frac{-1}{3} = m_1$

$\frac{4}{4}$

$y - 2 = m(x + 3)$

$y - 2 = \frac{-1}{3}(x + 3)$

$y - 2 = \frac{-1}{3}x - 1$ or

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

$\frac{-1}{3}x - y + 1 = 0$

c) verify the identity: $\sec 2\theta = \frac{\sec^2 \theta}{2 - \sec^2 \theta}$

$$\frac{1}{\cos^2 \theta} = \frac{1}{2 - \frac{1}{\cos^2 \theta}}$$

$$\frac{1}{\cos^2 \theta} = \frac{1}{\frac{2\cos^2 \theta - 1}{\cos^2 \theta}}$$

$$\frac{1}{\cos^2 \theta} = \frac{\cos^2 \theta}{2\cos^2 \theta - 1}$$

$$\frac{1}{\cos^2 \theta} = \frac{1}{\cos^2 \theta} \cdot \frac{\cos^2 \theta}{2\cos^2 \theta - 1}$$

$$\frac{1}{\cos^2 \theta} = \frac{1}{2\cos^2 \theta - 1}$$

$$\frac{1}{\cos^2 \theta} = \frac{1}{2\cos^2 \theta - 1}$$

$$\frac{\sec^2 \theta}{2 - \sec^2 \theta} = \sec 2\theta \text{ R.H.S.}$$

$$\frac{\sec^2 \theta}{2 - \sec^2 \theta} = \sec 2\theta \text{ R.H.S.}$$

Question (2): (8+4+3=15 marks)

a) Choose the correct answer

1) $|-2.7| = \dots -2 \dots$

(a) -3 , (b) -2 , (c) -7

2) $f(x) = 3x^3 - 4x^4$ is ... function (a) even , (b) odd , (c) neither even nor odd

3) The set of solution of the inequality $x^2 - 4x + 3 > 0$ is

(a) $(-\infty, 1) \cup (3, \infty)$, (b) $[1, 3]$, (c) $(1, 3)$

4) If $S = \frac{4\pi}{3}$, $r = 4 \text{ cm}$ then $\theta = \dots \dots \dots$ (a) $\frac{2\pi}{3}$, (b) $\frac{4\pi}{3}$, (c) 3π

$\frac{8}{8}$

b) Let $f(x) = \sqrt{x^2 - 4}$, find D_f and sketch the graph of f

$x^2 - 4 \geq 0$

$x^2 \geq 4$

$|x| \geq 2$

$\frac{4}{u}$

$x \geq 2 \text{ or } x \leq -2$

$[2, \infty) \cup (-\infty, -2]$

x	-4	-3	-2	2	3	4
$f(x)$	$\sqrt{12}$	$\sqrt{5}$	0	0	$\sqrt{5}$	$\sqrt{12}$

$\sqrt{(-4)^2 - 4} = \sqrt{16 - 4} = \sqrt{12}$

$\sqrt{(-3)^2 - 4} = \sqrt{9 - 4} = \sqrt{5}$

$\sqrt{(-2)^2 - 4} = \sqrt{4 - 4} = 0$

$\sqrt{(2)^2 - 4} = \sqrt{4 - 4} = 0$

$\sqrt{(3)^2 - 4} = \sqrt{9 - 4} = \sqrt{5}$

$\sqrt{(4)^2 - 4} = \sqrt{16 - 4} = \sqrt{12}$

c) Solve: $\log(\log x) + \log((\log x^3) - 2) = 0$

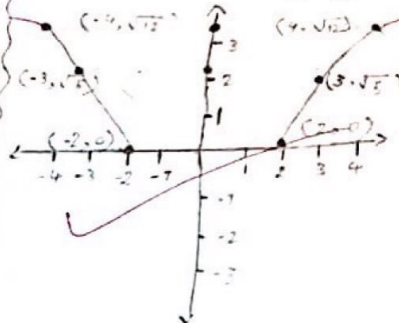
$\log(\log x)(\log x^3) - 2 = 0$

$x \cdot x^3$

$x^4 = 2$

$\log_x 2 = 4$

$\frac{0}{3}$



Question (2): (8+4+3=15 marks)

a) Choose the correct answer

1) $[-2.7] = \underline{-2}$

(a) -3, (b) -2, (c) -7

2) $f(x) = 3x^3 - 4x$ is ~~an~~ a function (a) even, (b) odd, (c) neither even nor odd

3) The set of solution of the inequality $x^2 - 4x + 3 > 0$ is $\underline{(1, 3)}$

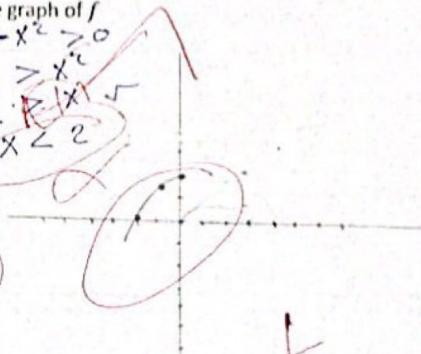
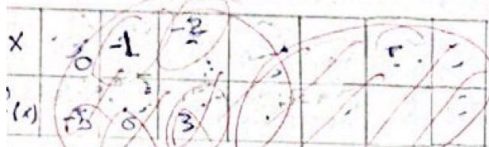
(a) $(-\infty, 1) \cup (3, \infty)$, (b) $[1, 3]$, (c) $(1, 3)$

4) If $S = \frac{4\pi}{3}$, $r = 2$ cm then $\theta = \underline{\frac{2\pi}{3}}$ (a) $\frac{2\pi}{3}$, (b) $\frac{4\pi}{3}$, (c) 3π

b) Let $f(x) = \sqrt{x^2 - 4}$, find D_f and sketch the graph of f

$4 - x^2 > 0$
 $x^2 < 4$
 $|x| < 2$
 $-2 < x < 2$

$4 - x^2 > 0$
 $4 > x^2$
 $2 > |x|$
 $-2 < x < 2$



c) Solve: $\log(\log x) + \log((\log x^3) - 2) = 0$

$\log x \cdot \log x^3 - 2$

$\log^2 - 2 = 0$

$\log^2 = 2$

$\log^2 = 10^2$

$\log^2 = 100$

$\frac{0}{3}$

$\frac{1}{4}$

$\frac{6}{8}$

Question (1): (8+4+3=15 marks)

a) Determine whether each of the following is true or false and correct the false statement:

- 1) If $2^{2x+3} = \frac{1}{8}$ then $x = \overset{-3}{(-2)}$ 4/8 (False.....)
- 2) The slope of a line passes through $A(-1,7), B(-3,2)$ is $\frac{5}{4}$ (True.....)
- 3) If $(f \circ g)(x) = \sqrt{x+1}$, then $(f \circ g)(3) = \textcircled{4}$ (False.....)
 $(f \circ g)(3) = \sqrt{3+1} = \sqrt{4} = 2$
- 4) The equation of the circle that satisfying the conditions : center in the second quadrant, tangent to both axis and radius 2 is $(x-2)^2 + (y-2)^2 = 2$ (True.....)

b) Find the equation of a line ($\vec{L}_1$) passes through the point $(1,2)$ that is perpendicular to the line $\vec{L}_2 : y = 2x + 1$.

~~2x+1~~

$$\begin{aligned} -y + 2x + 1 &= 0 \\ m_1 &= \frac{-a}{b} = \frac{-2}{-1} \\ m_1 &= 2 \\ m_2 &= \frac{-1}{2} = -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 2 &= \frac{1}{2}(x - 1) \\ 2(y - 2) &= x - 1 \\ 2y - 4 &= x - 1 \\ -2y + 4 &= -x + 1 \\ x - 2y + 6 &= 0 \end{aligned}$$

$$\frac{3\frac{1}{2}}{4}$$

c) verify the identity : $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{1 + \tan^2 \theta}{1 + \tan^2 \theta}$$

$$\begin{aligned} &= \frac{2 \tan \theta (1 + \tan^2 \theta)}{1 + \tan^2 \theta} \\ &= 2 \tan \theta \end{aligned}$$

$$\frac{0}{3}$$

a) Choose the correct answer

1) $[-1.9] = \dots\dots\dots$

(a) -1 , (b) -2 , (c) -9

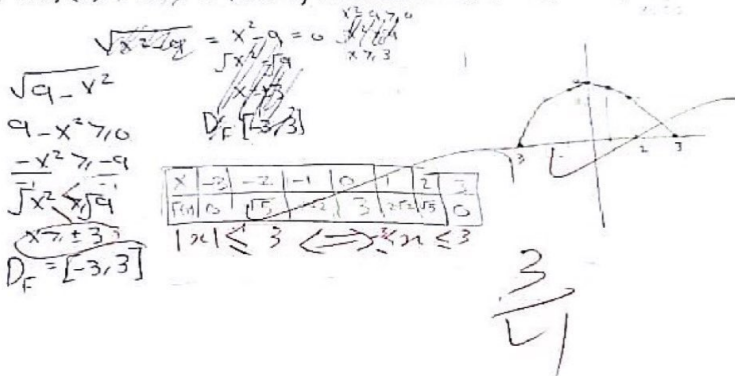
2) $f(x) = 9 - 5x^2$ is $\dots\dots\dots$ function (a) even, (b) odd, (c) neither

3) The set of solution of the inequality $x^2 - 3x - 4 < 0$ is $\dots\dots\dots$

(a) $(-\infty, -1) \cup (4, \infty)$, (b) $[-1, 4]$, (c) $(-1, 4)$

4) If $\theta = \frac{\pi}{2}$, $r = 2\text{ cm}$ then $A = \dots\dots\dots$ (a) $\frac{2\pi}{4}$, (b) $\frac{3\pi}{2}$, (c)

b) Let $f(x) = \sqrt{x^2 - 9}$, find D_f and sketch the graph of f



c) Solve: $\log(\log x) + \log((\log x^2) + 1) = 0$

$\log + \log x \cdot \log + 2\log x + 1 = 0$

$x(2x+1) = 0$

$2x^2 + x = 0$

$x = \frac{1}{2}$
 3

$$\log(\log x) + \log((\log x^2) + 1) = 0$$

$$\log(\log x) + \log(2\log x + 1) = 0$$

$$\text{Let } \log x = y$$

$$\log y + \log(2y + 1) = 0$$

$$\log(y(2y + 1)) = 0$$

$$y(2y + 1) = 10^0 = 1$$

$$2y^2 + 4 = 1$$

$$2y^2 + y - 1 = 0$$

$$a = 2 \quad b = 1 \quad c = -1$$

$$y = \frac{-1 \pm \sqrt{1 + 8}}{4} \Rightarrow \frac{-1 \pm \sqrt{12}}{4}$$

$$= \frac{-1 \pm \sqrt{4 \cdot 3}}{4} \Rightarrow \frac{-1 \pm 2\sqrt{3}}{4}$$

$$y = \frac{-1 + 2\sqrt{3}}{4} \Rightarrow \log x = \frac{-1 + 2\sqrt{3}}{4}$$

$$0.61 \dots$$

$$x = 10$$

Test in Differential Calculus

Time: 1 hour and 30 min

B

28

30

Name: C

Department: IT

Question (1): (8+4+3=15 marks)

a) Determine whether each of the following is true or false and correct the false statement:

1) If $7^{2x} \log_7 4 = 64$ then $x = 2$ $X = \frac{3}{2}$ (False...)

2) If $A(2,1), B(6,-2)$ is $d(A,B) = 5$ (True...)

3) If $f(x) = -\sqrt{5}$ then $D_f = \mathbb{R}/\{-\sqrt{5}\}$ $D_f = \mathbb{R} \rightarrow$ it's $\mathbb{R}$ (False...)

4) If the center of equation of the circle $x^2 + y^2 + 4x - 6y + 4 = 0$ is $(-2, -3)$ (False...)

b) If $L_1: 3x - 6y + 8 = 0$, find the slope and y-intercept of the line L_1 .

Solution: $3x - 6y + 8 = 0 \Rightarrow 3x + 8 = 6y$

$3x - 6y + 8$

$6y = 3x + 8 \div 6$

$m = \frac{-a}{b} = \frac{-3}{-6} = \frac{1}{2}$

$\frac{6y}{6} = \frac{3x}{6} + \frac{8}{6}$

y-intercept = $\left(\frac{c}{b}\right) = \frac{-8}{-6}$

$y = \frac{1}{2}x + \frac{4}{3}$

y-intercept = $\frac{-8}{-6} = \frac{4}{3}$

$m = \frac{1}{2}$

y-intercept = $\frac{4}{3}$

c) verify the identity: $\frac{\sin^2 \theta}{\cos \theta} = \sec \theta - \cos \theta$

Solution

$\frac{\sin^2 \theta}{\cos \theta} = \sec \theta - \cos \theta$

$\sin^2 \theta + \cos^2 \theta = 1$

$\sin^2 \theta = 1 - \cos^2 \theta$

$\frac{1 - \cos^2 \theta}{\cos \theta} \div \cos \theta$

$\frac{1}{\cos \theta} = \sec \theta$

$\frac{1 - \cos^2 \theta}{\cos \theta}$

$\frac{1}{\cos \theta} - \frac{\cos^2 \theta}{\cos \theta}$

$\sec \theta - \cos \theta$

Question (2): (8+4+3=15 marks)

a) Choose the correct answer

1) $|-1.7| = \underline{-2}$

(a) -1, (b) -2, (c) -7

2) $f(x) = 2x^2 - 3x + 4$ is (c) function (a) even, (b) odd, (c) neither even nor odd

3) The set of solution of the inequality $\frac{x-3}{x-1} \leq 0$ is (1, 3]

(a) $(-\infty, 1) \cup [3, \infty)$, (b) $[1, 3]$, (c) $(1, 3)$

4) If $\theta = 240^\circ$, the equation of a circle is $(x-1)^2 + (y+2)^2 = 9$ then

$S = \dots \underline{4\pi} \dots$ (a) 2π , (b) $\frac{4\pi}{3}$, (c) 4π

$S = r\theta \Rightarrow 3 \times \frac{4\pi}{3} = 4\pi$

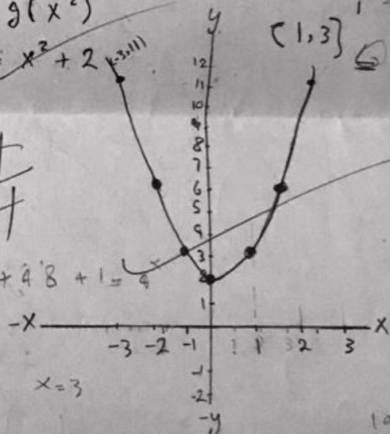
b) Let $f(x) = x^2$, $g(x) = x+2$, find $(g \circ f)(x)$ and sketch the graph of $g \circ f$

Solution $f(x) = x^2$, $g(x) = x+2$

$(g \circ f)(x) = g(f(x)) \Rightarrow g(x^2)$

x	-3	-2	-1	0	1	2	3
f(x)	9	4	1	0	1	4	9

$(g \circ f)(x) = x^2 + 2$
 $D_{g \circ f} = \mathbb{R}$



$\log_2(2^x + 48) + 1 = x$
 $\log_2(2^x + 48) = x - 1$

$2^x + 48 = 2^{x-1}$
 $2^x + 48 = 2^{x-1}$

c) Solve the equation: $\log_2(2^x + 48) + 1 = x$

$\log_2(2^x + 48) = x - 1$ $x = 4$

$\log_2(2^x + 48) = x$

$\log(2^x + 48) = x \log 2$

$\log(2^x + 48) = \log 4^x$

$x \log 2 = \log_2(2^x + 48)$

$2^x + 48 = 4^x$

$2^x = 4^x - 48$

$x = 4$

$2^x + 48 = 2^{2x}$

$2^x - 2^{1x} + 48 = 0$

$2^x - (2^2 \cdot 2^x) + 48 = 0$

$$\log_4 (2^x + 48) + 1 = x$$

$$\log_4 (2^x + 48) = x - 1$$

$$2^x + 48 = 4^{x-1}$$

$$2^x + 48 = 4 \cdot 4^{x-2}$$

$$2^x + 48 = (2^2)^{x-1}$$

$$2^x + 48 = (2^x)^2$$

$$(x+1) \times 4 \cdot 2^x + 192 = (2^x)^2$$

$$(2^x)^2 - 4 \cdot (2^x) - 192 = 0 \quad (\text{Let } u = 2^x)$$

$$u^2 - 4u - 192 = 0$$

$$(u - 16)(u + 12) = 0$$

$$u = 16 \quad \text{or} \quad u = -12$$

$$\Rightarrow 2^x = 16 \quad \text{or} \quad 2^x = -12$$

$$\Rightarrow 2^x = 2^4$$

$$x = 4$$

السالب
مرفوض
خارج النطاق

Name: _____

Department: IT

Q1) a) Determine whether each of the following is true or false:

9 1/2

$$1) \lim_{x \rightarrow 1} (x^2 - 3x + 4) = 8$$

$$2) \cot \frac{7\pi}{4} = 1$$

$$3) \text{ If } 6^{2x \log_6 5} = 625 \text{ then } x = 2$$

4 1/2

4 1/2

(...false...)

(...false...)

(...True...)

b) 1) Given the ellipse equation. find the eccentricity, the coordinates of the vertices, the foci, the equations of the directrices, and sketch the graph of $\frac{x^2}{9} + \frac{y^2}{2} = 1$

$$\frac{x^2}{9} + \frac{y^2}{2} = 1, \quad a^2 = 9 \Rightarrow a = 3, \quad b^2 = 2 \Rightarrow b = \sqrt{2}$$

$$c^2 = a^2 - b^2 \Rightarrow 9 - 2 \Rightarrow 7 \Rightarrow c^2 = 7 \Rightarrow c = \sqrt{7}$$

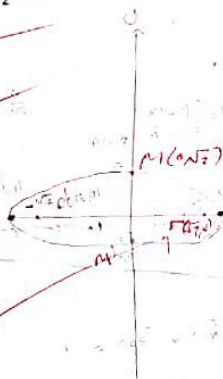
$$\text{eccentricity} = e = \frac{c}{a} = \frac{\sqrt{7}}{3}$$

$$\text{Vertices: } V(\pm a, 0) \Rightarrow V(\pm 3, 0)$$

$$\text{Foci: } F(\pm c, 0) \Rightarrow F(\pm \sqrt{7}, 0)$$

equations of the directrices:

$$X = \pm \frac{a^2}{c} = \pm \frac{9}{\sqrt{7}}$$



$$2) \text{ Show that } \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta} = \frac{2 \sin \theta \cos \theta}{1 - 2 \sin^2 \theta} = \frac{2 \sin \theta \cos \theta}{\sin^2 \theta + \cos^2 \theta - 2 \sin^2 \theta}$$

$$= \frac{2 \sin \theta \cos \theta}{\sin^2 \theta + \cos^2 \theta - 2 \sin^2 \theta} = \frac{2 \sin \theta \cos \theta}{1 - \cos^2 \theta + 2 \cos^2 \theta - 1}$$

$$1 \frac{1/2}{2}$$

Q.2) a) Choose the correct answer

- 1) If $\sec \theta = \frac{-13}{12}$ and $\theta \in III$, then $\cot \theta = \frac{12}{5}$ (a) $\frac{13}{12}$, (b) $\frac{12}{5}$, (c) $\frac{12}{13}$
- 2) If $\frac{1}{9} = 3^{2b-4}$, then $b = \dots$ (a) 2, (b) 1, (c) $\frac{1}{2}$
- 3) $\lim_{x \rightarrow 0} \frac{\sin(-3x)}{4x} = \frac{-3}{4}$ (a) -2, (b) $\frac{3}{4}$, (c) $-\frac{3}{4}$

b) 1) Evaluate $\lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{x-25}$ (if they exist)

$$\lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{x-25} = \lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{\sqrt{x}^2-5^2} = \lim_{x \rightarrow 25} \frac{(\sqrt{x}-5)}{(\sqrt{x}-5)(\sqrt{x}+5)} = \lim_{x \rightarrow 25} \frac{1}{\sqrt{x}+5}$$

$$= \frac{1}{\sqrt{25}+5} = \frac{1}{5+5} = \frac{1}{10}$$

2) Find $\log_{100} 25 + \log_{100} 40$ (without the calculator)

$$\log_{100} 25 + \log_{100} 40 = \log_{100} (25 \cdot 40)$$

$$= \log_{100} 1000 = \frac{\log_{10} 1000}{\log_{10} 100} = \frac{\log_{10} 10^3}{\log_{10} 10^2} = \frac{3}{2} = 1.5$$

Let $x = \log_{100} 1000$

$$1000 = 100^x$$

$$10^3 = 10^{2x}$$

$$3 = 2x$$

$$x = \frac{3}{2}$$

Name: _____

Department: IT

Excellent

Q.1) a) Determine whether each of the following is true or false:

- 1) $\frac{7\pi}{6} \text{ rad} = 210^\circ$ 6/6 ✓ (true.....)
- 2) Let $A(1,2), B(-3,5)$ then $d(A,B) = 5$ ✓ (true.....)
- 3) If $(f \circ g)(x) = \sqrt{x^2 + 1}$, then $(f \circ g)(3) = 2$ 6/6 ✓ (False.....)
- 4) The center of the circle $(x-1)^2 + (y+1)^2 = 9$ is $(-1,1)$ ✓ (False.....)

- b) 1) Find the equation of a line passes through the point $(1,2)$ that is parallel to the line $4x + 2y = 3$.

∴ line parallel the slope (m) ∴ then $m_1 = m_2$

$$m = \frac{-a}{b} = \frac{-4}{2} = -2 \quad \text{or } 4x + 2y = 3 \Rightarrow 2y = -4x + 3 \Rightarrow y = -2x + \frac{3}{2}$$

$$m = -2$$

the equation of

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -2(x - 1)$$

$$y - 2 = -2x + 2$$

$$2x + y = 2 + 2$$

$$2x + y = 4 \Rightarrow 2x + y - 4 = 0$$

- 2) Find the equation of a line passes through the point $(1,2)$ that is perpendicular to the line $4x + 2y = 3$.

∴ line perpendicular the slope (m) ∴ then $m_1 \cdot m_2 = -1$

$$m = \frac{-a}{b} = \frac{-4}{2} = -2$$

$$m_1 = -2$$

$$m_1 \cdot m_2 = -1 \Rightarrow -2 \cdot m_2 = -1 \Rightarrow m_2 = \frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{1}{2}(x - 1)$$

$$y - 2 = \frac{x}{2} - \frac{1}{2}$$

$$y - \frac{x}{2} = 2 - \frac{1}{2}$$

$$2y - x = 4 - 1$$

$$2y - x = 3$$

$$\frac{4}{4}$$

Q2) a) Choose the correct answer

9 1/2

1) $|-0.1| = \dots -1 \dots$

(a) 0, (b) -1, (c) 1

2) $|3 - \pi| = \dots \pi - 3 \dots$

(a) $\pi - 3$

(b) $\pi + 3$

(c) $3 - \pi$

3) The set of solution of the inequality $\frac{x+3}{x-2} \leq 0$ is $[-3, 2)$

(a) $(-\infty, -3) \cup (2, \infty)$, (b) $[-3, 2]$, (c) $[-3, 2)$

6/6

1) Let $f(x) = \frac{1}{x^2+4}$, find D_f and Range of f

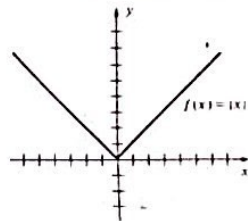
2) Let $f(x) = 2x^3 + x^2$, determine whether f is even, odd or neither even nor odd

3) given the graph of the function $f(x) = |x|$, use that

to sketch the graph $f(x) = |x-2| + 1$ (shifts)

$D_f = \mathbb{R}$
 $\therefore D_f = \mathbb{R}$

$\frac{1}{x^2+4} \geq 0$ $x^2+4 = 0$
 $x^2 = -4$
no real solution



2) Range: let $y = \frac{1}{x^2+4} \Rightarrow y(x^2+4) = 1 \Rightarrow$

$\Rightarrow yx^2 + 4y = 1 \Rightarrow yx^2 = -4y + 1$

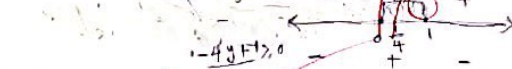
$\Rightarrow x^2 = \frac{-4y+1}{y}$

$|x| = \sqrt{\frac{-4y+1}{y}} > 0$

$\frac{-4y+1}{y} > 0$

$-4y+1 = 0 \Rightarrow -4y = -1 \Rightarrow y = \frac{-1}{-4} = y = \frac{1}{4}$

$y = 0$

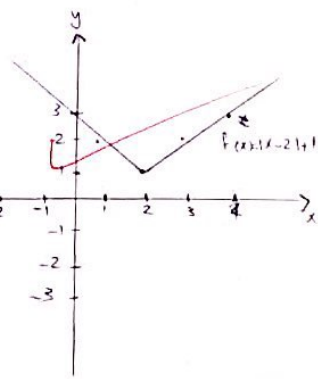


Range = $(0, \frac{1}{4}]$

$P(x) = \frac{1}{x^2+4}$
 $f(-x) = \frac{1}{(-x)^2+4} = \frac{1}{x^2+4}$ even

السؤال آخر

3 1/2 / 4



نماذج الامتحانات النهائية

تجميع :

نور الجفري & فاطمة عاشور



HADHRAMOUT UNIVERSITY
COLLEGE OF COMPUTERS & INFORMATION TECHNOLOGY
FINAL EXAMINATION



Academic year: 2019/2020

Day and Date: Sunday 15/12/2019

Examiner: Rafat Khalid Bin Talib

Time allowed: 2 Hours & 30 Minutes

Exam Semester: First

Level: First (قبول عام - منقحة خاصة)

Department: IT & CS

Subject: Differential Calculus

Answer only four of the following questions. The first one is obligatory:

(اجب عن أربعة أسئلة فقط على أن يكون السؤال الأول اجبارياً)

Question (1): (15+10=25 marks)

(a) Choose the correct answer:

1) $[6.4] = \dots$ a) 6, b) 7, c) 4

2) Since $\lim_{x \rightarrow 2^+} \frac{1}{(x-2)^3} = \infty$ then the vertical asymptote is $x = 2$

3) If $f(x) = \log_a x$, then $D_f = \dots$ a) $\mathbb{Z}$, b) $\mathbb{R}$, c) $(0, \infty)$

4) The critical number of $f(x) = x^2$ is 0 a) 1, b) 0, c) -1

5) The function $f(x) = 3x^3 - 4x$ is odd
a) Even, b) Odd, c) Neither even nor odd

(b) Given the ellipse equation. find the eccentricity, the coordinates of the vertices, the foci, the equations of the directrices, and sketch the graph $\frac{x^2}{9} + \frac{y^2}{25} = 1$

Question (2): (5+10+10=25 marks)

(a) Show that $\tan^2 \theta (1 + \cot^2 \theta) = \sec^2 \theta$

(b) Find $\frac{dy}{dx}$ for each of the following

1) $y = \sqrt{x^3 + x + 1}$, 2) $x + \sin y = xy$

(c) If $f(x) = x^3 + x^2 - 5x - 5$, find the intervals on which f is increasing and the intervals on which f is decreasing, and Find the local extrema of f

Question (3): (7+10+8=25 marks)(a) Find $\frac{d^2y}{dx^2}$ (the second derivative) of $y = \sec x$

(b) Solve each of the following

1) $\ln(y+1) = \ln(y-2) + \ln 2$, 2) $|x-3| - 2 = -5$

(c) Prove that the function f defined by $f(x) = x^3 - 8x - 5$ satisfies the hypotheses of the Mean Value Theorem on the interval $[1,4]$ and find all numbers c in the interval $(1,4)$ that satisfies the conclusion of the theorem.Question (4): (5+8+12=25 marks)(a) Suppose $f(x) = \begin{cases} cx^2 - 3 & , \text{ if } x \leq 2 \\ cx + 2 & , \text{ if } x > 2 \end{cases}$ Find a value of c such that f is continuous at $x = 2$.

(b) Find the equation of the

1) Line that passing through $(1,0)$ and $m = -2$

2) circle that tangent to both axes, center in the second quadrant, radius 4

(c) If $f(x) = 3x^2 - 12x + 8$ find1) Find the equations of the tangent and the normal line at $(3,1)$

2) The point on the graph at which the tangent line is a horizontal

Question (5): (15+10=25 marks)

(a) Find the limits (if they exists)

1) $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$, 2) $\lim_{x \rightarrow -\infty} \frac{x^2 - 3x + 7}{x^3 + 4x - 9}$, 3) $\lim_{x \rightarrow 0} \frac{\log_2 4^x}{\tan x}$

(b) If $f(x) = 12 + 2x^2 - x^4$, use the Second Derivative Test to find local maxima and minima of f . Discuss concavity, find the points of inflection.

$$6x - 12 = 0$$

$$6x = 12$$

$$x = 2$$

$$12 - 24 + 8$$

$$-12 + 8$$

$$-4$$

$$3 - 12 + 8$$

$$3 - 4$$

$$-1$$

Good Luck

$$(x-h)^2 + (y-k)^2 = r^2$$



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Question (1): (15+10=25 marks)

(a) Choose the correct answer:

1) $[6.4] = \dots$ a) 6 , b) 7 , c) 4

2) Since $\lim_{x \rightarrow 2^+} \frac{1}{(x-2)^3} = \infty$ then the vertical asymptote is $x = 2$

3) If $f(x) = \log_a x$, then $D_f = \dots$ a) $\mathbb{Z}$, b) $\mathbb{R}$, c) $(0, \infty)$

4) The critical number of $f(x) = x^2$ is 0 a) 1 , b) 0 , c) -1

5) The function $f(x) = 3x^3 - 4x$ is odd a) Even , b) Odd , c) Neither even nor odd

(b) Given the ellipse equation. find the eccentricity, the coordinates of the vertices, the foci, the equations of the directrices, and sketch the graph $\frac{x^2}{9} + \frac{y^2}{25} = 1$

Question (2): (5+10+10=25 marks)

(a) Show that $\tan^2 \theta (1 + \cot^2 \theta) = \sec^2 \theta$

(b) Find $\frac{dy}{dx}$ for each of the following

1) $y = \sqrt{x^3 + x + 1}$, 2) $x + \sin y = xy$

(c) If $f(x) = x^3 + x^2 - 5x - 5$, find the intervals on which f is increasing and the intervals on which f is decreasing, and Find the local extrema of f

Question (3): (7+10+8=25 marks)

(a) Find $\frac{d^2y}{dx^2}$ (the second derivative) of $y = \sec x$

(b) Solve each of the following

1) $\ln(y+1) = \ln(y-2) + \ln 2$, 2) $|x-3| - 2 = -5$

(c) Prove that the function f defined by $f(x) = x^3 - 8x - 5$ satisfies the hypotheses of the Mean Value Theorem on the interval $[1,4]$ and find all numbers c in the interval $(1,4)$ that satisfies the conclusion of the theorem.

Question (4): (5+8+12=25 marks)

(a) Suppose $f(x) = \begin{cases} cx^2 - 3 & , \text{ if } x \leq 2 \\ cx + 2 & , \text{ if } x > 2 \end{cases}$

Find a value of c such that f is continuous at $x = 2$.

(b) Find the equation of the

1) Line that passing through $(1,0)$ and $m = -2$

2) circle that tangent to both axes, center in the second quadrant, radius 4

(c) If $f(x) = 3x^2 - 12x + 8$ find

1) Find the equations of the tangent and the normal line at $(3,1)$

2) The point on the graph at which the tangent line is a horizontal

Question (5): (15+10=25 marks)

(a) Find the limits (if they exists)

1) $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$, 2) $\lim_{x \rightarrow -\infty} \frac{x^2 - 3x + 7}{x^3 + 4x - 9}$, 3) $\lim_{x \rightarrow 0} \frac{\log_2 4^x}{\tan x}$

(b) If $f(x) = 12 + 2x^2 - x^4$, use the Second Derivative Test to find local maxima and minima of f . Discuss concavity, find the points of inflection.

$$6x - 12 = 0$$

$$6x = 12$$

$$x = 2$$

$$12 - 24 + 8$$

$$= -12 + 8$$

$$= -4$$

$$\begin{aligned} 3 - 12 + 8 \\ 3 - 4 \\ -1 \end{aligned}$$

Good Luck



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 (أجب عن أربعة أسئلة فقط على أن يكون السؤال الأول إجبارياً)

Question (1): (15+10=25 marks)

(a) Choose the correct answer:

- 1) $[7.3] = \dots$ a) 7 , b) 8 , c) -7
- 2) Since $\lim_{x \rightarrow \infty} \frac{2x^2 - 5}{3x^2 + x + 2} = \frac{2}{3}$ then the horizontal asymptote is $\dots$
 a) $y = 0$, b) $y = \frac{2}{3}$, c) $y = \frac{-5}{2}$
- 3) If $f(x) = |x|$, then $D_f = \dots$ a) $\mathbb{Z}$, b) $\mathbb{R}$, c) $(0, \infty)$
- 4) The critical number of $f(x) = x^2 - 4x$ is $\dots$ a) 4 , b) 0 , c) 2
- 5) The function $f(x) = x^2 + 2$ is $\dots$
 a) Even , b) Odd , c) Neither even nor odd

(b) Find the vertices , foci and asymptotes of the hyperbola given by the equations , the equations of the directrices , and sketch the graph of each equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$

Question (2): (5+10+10=25 marks)

(a) Show that $\frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$

(b) Find $\frac{dy}{dx}$ for each of the following

- 1) $y = (x^3 + 8)^5$, 2) $x^2y + 3y^2 = 6$

(c) If $f(x) = x^3 + x^2 - 5x - 5$, find the intervals on which f is increasing and the intervals on which f is decreasing, and Find the local extrema of f

Question (3): (7+10+8=25 marks)

(a) Find $\frac{d^2y}{dx^2}$ (the second derivative) of $y = \tan x$

(b) Solve each of the following

1) $\ln(3x+8) = \ln(2x+2) + \ln(x-2)$, 2) $x^2 - 7x + 10 > 0$

(c) Prove that the function f defined by $f(x) = 4x^2 - 20x + 29$ satisfies the hypotheses of the Rolle's Theorem on the interval $[1,4]$ and find all numbers c in the interval $(1,4)$ such that $f'(c) = 0$

Question (4): (5+8+12=25 marks)

(a) If $f(x) = \begin{cases} x+3 & \text{if } x < 2 \\ x^2+1 & \text{if } x \geq 2 \end{cases}$ discuss the continuity of $f(x)$ at $x = 2$.

(b) Find the equation of the

1) circle passing through $(2,3)$ and $(4,5)$

2) circle tangent to both axes, center in the first quadrant, radius 3

(c) If $f(x) = 3x^2 - 12x + 8$ find

1) Find the equations of the tangent and the normal line at $(1,0)$

2) The point on the graph at which the tangent line is a horizontal

Question (5): (15+10=25 marks)

(a) Find the limits (if they exist)

1) $\lim_{x \rightarrow 1} \frac{x^2+x-2}{x^2-x}$, 2) $\lim_{x \rightarrow \infty} \frac{7x^2+x+11}{4-x}$, 3) $\lim_{x \rightarrow 0} \frac{\sin^2 3x}{3 \ln \cos x}$

(b) If $f(x) = x^3 + x^2 - 5x - 5$, use the Second Derivative Test to find local maxima and minima of f . Discuss concavity, find the points of inflection.

$3x^2 - 7x - 5$
 $(6x + 2)$

$\frac{2x^3 + 2x^2}{x^2} = 2x + 2 = \frac{2x}{2} + 2 = \frac{-1}{3}$

$[-\infty, -\frac{1}{3}] \cup [\frac{1}{3}, \infty]$

Good Luck



Academic year: 2017 - 2018

Day and Date: Wednesday / 14 February - 2018

Examiner: Dr. Ahmed Kherd

Time allowed: 2 Hours.

Exam Semester: First
Level: First (صباحي + مسائي)
Department: CS + IT
Subject: Differential Calculus

Answer any four from the following questions:

Question one:

- 1) Solve the inequality

$$-5 < \frac{4-3x}{2} < 1$$

- 2) Prove that the triangle with vertices $A(-1, -3)$, $B(6, 1)$ and $C(2, -5)$ are right triangle.

- 3) Find the equation of the line through the point $(5, -7)$ that is parallel to the line

$$6x + 3y - 4 = 0.$$

Question two:

- 1) Let $f(x) = x - 2$ and $g(x) = 5x + \sqrt{x}$ then find

i. $(f \circ g)(x)$

ii. $(g \circ f)(x)$

- 2) Use the definition of limit to show that

$$\lim_{x \rightarrow 4} \frac{1}{2}(3x - 1) = \frac{11}{2}$$

- 3) Is the limit $\lim_{x \rightarrow 4} \frac{|x-4|}{x-4}$ exist?

Question three:

- 1) Use the definition of derivative to find the derivative of $f(x) = \sqrt{x}$.

2) Find $\lim_{x \rightarrow 4} \frac{x-2}{x^2-6}$.

- 3) Prove that $D_x[\cot x] = -\csc^2 x$.

4) Find y' if $y = \frac{1}{(4x^2 + 6x - 7)^7}$.

Question four:

- 1) Let $y = \sin^3(4x)$. Then show that $y' = 12\sin^2(4x)\cos(4x)$.
- 2) Let $g(z) = \left(z^2 - \frac{1}{z^2}\right)^6$. Then find $D_z[g(z)]$.
- 3) If $y^4 + 3y - 4x^3 = 5x - 1$. Then find the implicit derivative of y ?

Question five:

- 1) Plot the graph of the function $f(x) = |x - 1|$.
- 2) Use the sandwich theorem to prove that

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0.$$

- 3) Find the slope of tangent line to the graph of $y^4 + 3y - 4x^3 = 5x - 1$ at the point $P(1, -2)$.

Good Luck

